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Developing WSN/BIM-Based Environmental Monitoring Management System for Parking Garages in Smart Cities

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Abstract: In recent years, there have been many serious problems with air pollution (such as hazardous gases) in environmental management for parking garages of cities. These problems should be continuously improved and enhanced for smart cities. Building information modeling (BIM) has been revolutionary in construction industry applications; it integrates different kinds of construction information into a 3D digital model and can be applied in all stages of a project lifecycle. Furthermore, the wireless sensor network (WSN) is an important key technology in Internet of Things (IoT) development. In order to improve the performance of environmental monitoring management of parking garages in smart cities, this study proposes a planning approach for integration of WSN and BIM technologies and develops an advanced monitoring and control system for underground parking garage environment monitoring, adopting WSN and BIM technologies for digital twin (DT) application of Industry 4.0 (I4.0). Many WSN nodes with sensing and control capabilities were placed in a parking garage to collect carbon monoxide (CO), temperature, and humidity conditions in real time. All the collected information was transferred by a WSN self-constituted network, and the different location conditions were integrated and dynamically displayed on corresponding spatial components of a 3D BIM model. The proposed system was applied in a case study of a parking garage in Taipei to verify the system and demonstrate its effectiveness in practice. The results demonstrate that the proposed system is a visually effective monitoring system for environmental monitoring management. Finally, this study proposes a conclusion, suggestions, benefits, and limitations for further application. **DOI: 10.1061/(ASCE)ME.1943-5479.0000760.** *© 2020 American Society of Civil Engineers.*

Author keywords: Smart cities; Parking garages; Internet of Things (IoT); Wireless sensor network; Interoperability; Safety management; Building information modeling; Digital twins.

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Introduction

Rapid growth of traffic and vehicles makes air pollution an increasing concern in cities. There are many parking garages built in underground basements, which are closed spaces where the air is frequently polluted by vehicles. In garage management, people often focus attention on parking space optimization but ignore the air quality problems. During rush hour, high quantities of vehicles are moving in the closed space, constantly emitting exhaust gases. As the analysis displayed, the main composition of the polluting gas emitted by vehicles was CO, which is a colorless and odorless gas and is not easy to measure. According to the Journal of the American Medical Association (JAMA), 1,500 people die annually due to accidental CO gas exposure, and an additional 10,000 seek medical attention (Modic 2003). CO poisoning may be possible after inhaling only a 35-ppm concentration of CO gas, and if inhaling the hazardous gas continues, unconsciousness and death may result. Moreover, in many cases, garage air quality is maintained by forced air circulation or air conditioner units. However, long-term operation of this equipment will result in high electricity costs and energy waste; hence, a smart and proactive gas and environmental monitoring and control system is needed for maintaining air quality and people's health and safety.

The evolution progress of smart cities depends on the development of industry and technical levels. Industry 4.0 (I4.0) has focused on enhancing industry intellectualization, which integrates existing technologies and processes and then builds an adaptive, resource-efficient, intelligent production process. The Consortium II fact sheet (Consortium II 2013) defined I4.0 as "the integration of complex physical machinery and devices with networked sensors and software, used to predict, control, and plan for better business and societal outcomes. I4.0 facilitates interconnection and computerization into the traditional industry, which makes automatic and flexible adaptation of the production chain and provides new types of services and business models of interaction in the value chain" (Lu 2017).

During the modular structured smart factories of I4.0, a cyber-physical system (CPS) creates a virtual copy of the physical production and makes decentralized decisions (Antonio 2018). CPS is a mechanism that is controlled or monitored by computer-based algorithms, tightly integrated with the internet and its users (Monostori et al. 2016). Digital twin (DT) is another concept associated with I4.0; it is a dynamic virtual representation of a physical object or system across its lifecycle, using real-time data to enable understanding, learning, and reasoning (Bolton et al. 2018), and has a similar essence to CPS. Furthermore, DT has the potential to gather and analyze information at different scales, facilitate more efficient construction processes, and allow for informed decision-making, which will ultimately promote a safer environment at reduced costs and improved efficiency (Kan and Anumba 2019).

The two key factors of I4.0 are integration and interoperability (Chen et al 2008; Romero and Vernadat 2016). Integrating computer and network systems, I4.0 achieves seamless cooperation across

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organizations and industries. Interoperability is defined as "the ability of two systems to understand each other and to use functionality of one another," which represents the capability of two systems exchanging data and sharing information and knowledge (IDABC 2005).

Building information modeling (BIM) is a relatively new innovation for the construction and civil engineering industry (Tulubas Gokuc and Arditi 2017; Ma et al. 2018; Hosseini et al. 2018). BIM can not only enhance productivity in particular activities but can build system capabilities that enable new ways of working (Holmström et al. 2014). One of the characteristics of BIM technology is that it can be communicated by 3D illustration among participants. This characteristic is why BIM is selected as the main technology in the study. Also, wireless sensor networks (WSNs) are one of the important key technologies for Internet of Things (IoT) development. The sensing and control capabilities of WSN devices enable them to be placed in locations of the garage to collect carbon monoxide (CO), temperature, and humidity conditions for applications of DT.

In this study, a planning approach for integration of WSN and BIM technologies is proposed and a WSN/BIM-based environmental monitoring system is developed for underground parking garage environmental management. A smart WSN system integrated with environmental monitoring, hazardous gas detection, and risk status capability is set up in the garage to monitor the CO gas concentration and maintain environmental safety. Data on CO concentration, temperature, and humidity (environmental conditions) of the different locations are collected and visually represented on a digital building model in real time. The locational risk status is designed to be dynamically indicated by color, from blue to red, and the manager can master the overall safety and environmental conditions of the garage via the active model. The proposed system provides a presentation of the DT model of I4.0 and ubiquitous computing for smart city application. Moreover, in view of integration and interoperability, the WSN provides active information measured from the real world and enables the BIM model to be a real-time DT system and show the real status; on the other side, the BIM provides a digital structure so that the WSN data can be correctly embedded and show the dimensional relation of the occurring position for observation. The multifocus information integrated BIM model also demonstrates a real-time dimensional relation of risk priority and overall environmental information. The integrated system provides not only a solution for DT application but a meaningful, efficient system for safety management and decision-making in an emergency.

The proposed system can enhance the regular management and decision-making in a critical situation. Additionally, an intelligent function was designed that can remove the hazardous gas automatically when detecting critical conditions. The main purposes of this research include: (1) to propose a planning approach for the integration of WSN and BIM technologies for parking garages of smart cities, (2) to develop a WSN/BIM-based environmental monitoring system, and (3) to discuss the advantages and disadvantages of the developed system applied in the Taiwan case study.

Literature Review

Smart city refers to the use of various information technologies or innovative ideas to integrate urban systems and services to improve the efficiency of resource utilization and optimize urban management and services. The foundational concepts are instrumented, interconnected, and intelligent (Harrison et al. 2010). *Instrumented* refers to sources of near-real-time real-world data from both physical and virtual sensors; *interconnected* means the integration of those data into an enterprise computing platform and the communication of such information to the various city services; *intelligent* refers to the inclusion of complex analytics, modeling, optimization, and visualization in the operational business process to make better operational decisions.

With the onset of Industry 4.0, many advanced technologies have brought great potential applications for smart city environmental quality management. The DT is one of the main technologies associated with Industry 4.0 and is defined as "a unified system model that can coordinate architecture, mechanical, electrical, software, verification, and other discipline-specific models across the system lifecycle" (Bajaj et al. 2016). A DT can be used to simulate the physical asset or operation for various purposes. Many sensors with various functions can be adopted for understanding the real situation, exploiting a real-time synchronization of the sensed data originating from the field level, and deciding between a set of actions with the focus to orchestrate and execute the whole production system in an optimal way (Negri et al. 2017). WSNs are one of the core technologies for IoT solutions. A number of independent sensor nodes comprise a network that is able to monitor, communicate, and even judge and react to external circumstances (Cheung et al. 2018). With the advantages of small dimension, low price and power consumption, and ability in environmental detection and active networking, WSNs are widely applied for environmental monitoring, building intelligentization, smart city functions, etc. BIM provides digitalized and parameterized building information in a 3D representative model to enable professionals to efficiently plan, design, construct, and manage by visualization (Cerovsek 2011). The definition of BIM by the US National Institute of Building Sciences (NIBS) is "A digital representation of physical and functional characteristics of a facility" (NIBS 2019). In intelligent building and smart city application, BIM provides a powerful platform to integrate multifocus building information, such as asset spatial relations, regional conditions, and the simulation of asset movement and the construction process. A combined system using WSNs and BIM can establish a real-time active model that works as an application of a DT and provides efficient information for managers to make decisions. Moreover, the BIM model contains assets and spatial information, which provides a framework for the IoT data to be organized and analyzed, and the combined information is meaningful to the building operations and management (Haines 2016).

In environmental air quality research of parking garages, Papakonstantinou et al. (2003) examined ventilation effectiveness by computational and experimental investigation. They found that an efficient ventilation system will result in a better quality of air inside buildings, and reducing CO levels in occupational spaces is an important issue for short- and long-term health problems. Zhao et al. (2017) evaluated the seasonal patterns of particulate matter PM10, PM2.5, and PM1.0 in a naturally ventilated underground residential parking garage using field measurements collected at various times over a 1-year period. The results showed the PM10 and PM2.5 concentration exceeding the limit of the Chinese standard by eightfold during the spring, fall, and winter using natural ventilation. Kim et al. (2007) proposed a time-resolved measurement method, evaluated air toxin levels within an urban parking garage, and assessed the influence of vehicle activity and type on those levels. Carbon monoxide (CO), particle-bound polycyclic aromatic hydrocarbons (pPAH), and volatile organic compounds (VOCs) were measured and calculated. The influence of traffic was quantified using longitudinal models, and the pollutant coefficients provide an indication of the average contribution to air pollution from vehicle sources. This information demonstrates that the

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air quality of parking garages is an important issue in human health and safety.

Some studies have been done on the integration of WSN and BIM technologies. Tang et al. (2019a) comprehensively reviewed 97 papers from 14 architecture, engineering, and construction (AEC) related journals and databases over the last decade about the approach of BIM-IoT integration. Farghaly et al. (2018) explored the asset owner requirements from BIM in the operation and maintenance stage from an asset management (AM) perspective and output asset-consuming energy information management (ACE-IM) taxonomy to identify the required data and improve AM processes.

Cao et al. (2018) used the structural social capital perspective and a longitudinal data set on BIM implementation practices in the construction industry in Shanghai (from 2007 to 2015) to explore the structural characteristics of the project-based collaboration networks. Tang et al. (2019b) explored the direct relationship between BIM implementation and the performance of off-site manufacturing projects through their key influential factors. Awolusi et al. (2019) provided an evaluation of the potential applications of wearable sensing devices and IoT for the continuous collection, analysis, and monitoring of construction workers' safety metrics. Dave et al. (2018) developed a platform that integrates the built environment data with IoT sensors campuswide to provide information regarding energy usage, occupancy, and user comfort. Kan and Anumba (2019) presented a comprehensive, up-to-date literature review and provided critical analysis of existing research into DT applications. Fernández-Caramés et al. (2018) developed a system to allow for identifying and tracking the pipes of a ship through their construction cycle. Giel and Issa (2016) developed a framework for building-owner organizations to use in the assessment of their BIM competency and identify 66 critical factors about the evaluation of owners' BIM competency using the Delphi technique. Zhou et al. (2019) proposed a cyber-physical-system-based safety monitoring system to simulate and monitor the hoisting process to avoid the unsafe status of cranes in metro and underground construction and successfully implemented the system in Wuhan Metro's road tunnel. Gao et al. (2019) proposed a scalable data acquisition framework for the smart build of environment-smart buildings, smart communities, and smart cities that enables the utilization of the data housed in separate building systems for innovative IoT use.

Based on the literature surveyed, BIM-sensor technologies have been applied to facility automation and energy saving since 2007, propelling the progress of *building intelligence*. In recent decades, owing to advances in wireless sensor technologies, the relative research issues and applications have become more profound and diversified. Due to sensor size and price reduction, a large number of sensors can be utilized to collect a huge amount of information, which supports more extensive and detailed analysis of smart cities such as IoT and Internet of Vehicles (IoV) or the human/object interaction detection in virtual reality (VR) and augmented reality (AR) applications. In Industry 4.0, WSNs provide real-time information for DT models in the manufacturing and construction industries. The provided multifocus information enables the manager to understand the real situation of the object or system and respond to various changes accurately and quickly, thereby improving the operation of the system while enhancing efficiency and human safety.

Although there is some previous academic and practical research regarding integration of BIM and WSN technologies based on the literature review, few studies focus on the initial planning and considerations for WSN/BIM-based environmental monitoring management for parking garages in smart cities. Therefore, this study will focus on the aforementioned content for effective environmental monitoring management.

Methodology

It is difficult to measure air quality and detect the presence of hazardous gases in underground parking garages because gas concentrations and flow conditions respond dynamically to the quantity and movement of vehicles. This is compounded by CO gas being colorless and odorless and requiring a specific gas sensor to be measured. Moreover, the overall conditions at several locations are not easy to measure simultaneously by a single gas detector. With a lack of information about the status of each location and a smart control system, the manager cannot easily confirm where the high-concentration gas is coming from and has to keep the air conditioning equipment running for a long time to reduce the risk posed by the hazardous gases, leading to high electricity and maintenance costs. Efficient management of the overall garage conditions and the control of the air conditioning equipment will be a challenge.

The integrated DT becomes an important solution for smart cities. The purpose of DT is to provide a digital virtual illustration for the actual situation and enhance the dynamic change management for smart cities. Many related technologies can assist in the application of DT for smart cities. Specially, the assistance of BIM technology can provide visual 3D communication for DT. With the use of IoT technologies, much information can be acquired from the devices located in the selected areas. Using integrated BIM and IoT technologies, the 3D visual illustration can enhance the management of DT for smart cities (Fig. 1). The major purpose of this study is to propose an approach for DT integrated with BIM and IoT technologies for parking garages of smart cities. The following is a detailed description of this approach.

In this study, BIM models of DT are used to show different control statuses and illustrate where the identified problems are located. Therefore, it is important to remove unnecessary information from the final as-built model for a management-based BIM model. Usually, the performance of a BIM-based information system will be reduced if the BIM model is too detailed and complicated, so the degree of detail of BIM model development should be low for management. The suggested level of development (LOD) of the BIM model is LOD 200 or LOD 300 for suitable management. The aforementioned process at the initial stage of the operation phase can enhance the performance of a BIM-based information system for management purposes.

WSN is selected as the IoT solution for monitoring and recording the physical conditions of the environmental monitoring management in the study. With the use of WSN technologies, essential information can be acquired for monitoring management of devices located in the selected areas. It is necessary to identify the requirements of monitoring information acquisition initially. The monitoring information includes the monitoring contents and frequency. The monitoring content will influence all the plans for the environmental monitoring management. Therefore, it is essential to decide which monitoring content needs to be detected in the beginning. Furthermore, it is suggested to consider the setup quantity and location of the WSN devices for monitoring management. The reason is that the setup cost is too high if there are many WSN devices, but the performance of monitoring management will be ineffective if there are only a few WSN devices. Fig. 2 shows the workflow of integrated WSN and BIM technologies for parking garages of the smart cities in the study.

Regarding platform development for environmental monitoring management, a web-based system is selected as the DT monitoring system to allow all related participants to log in and refer to the monitoring information in order to make suitable decisions. Based on the aforementioned major requirement considerations, the DT monitoring system is developed via integration of a WSN and

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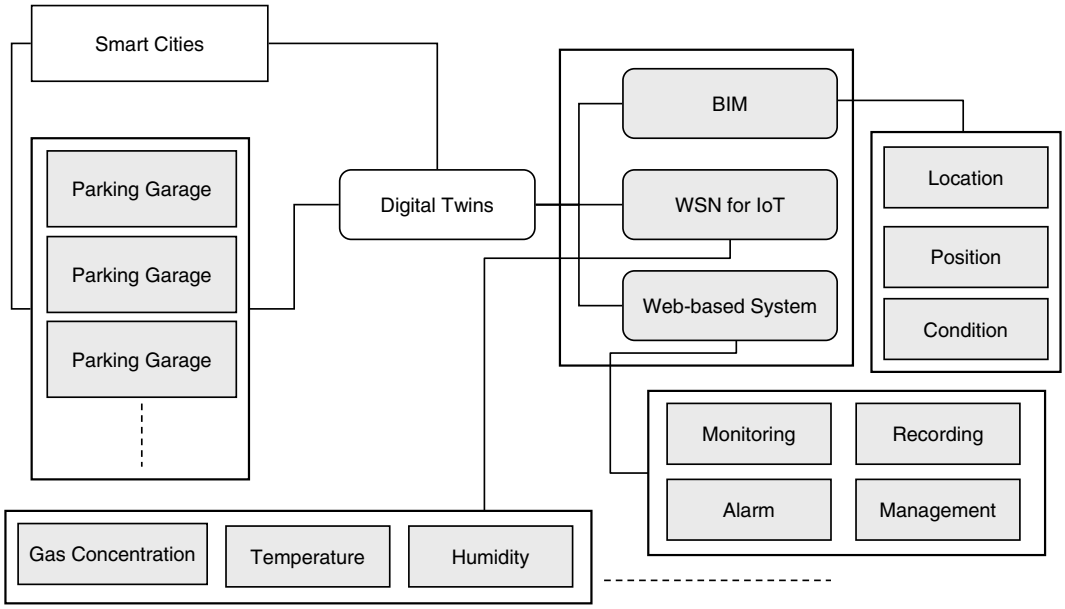


Fig. 1. Concept of proposed gas and environment management system for parking garages.

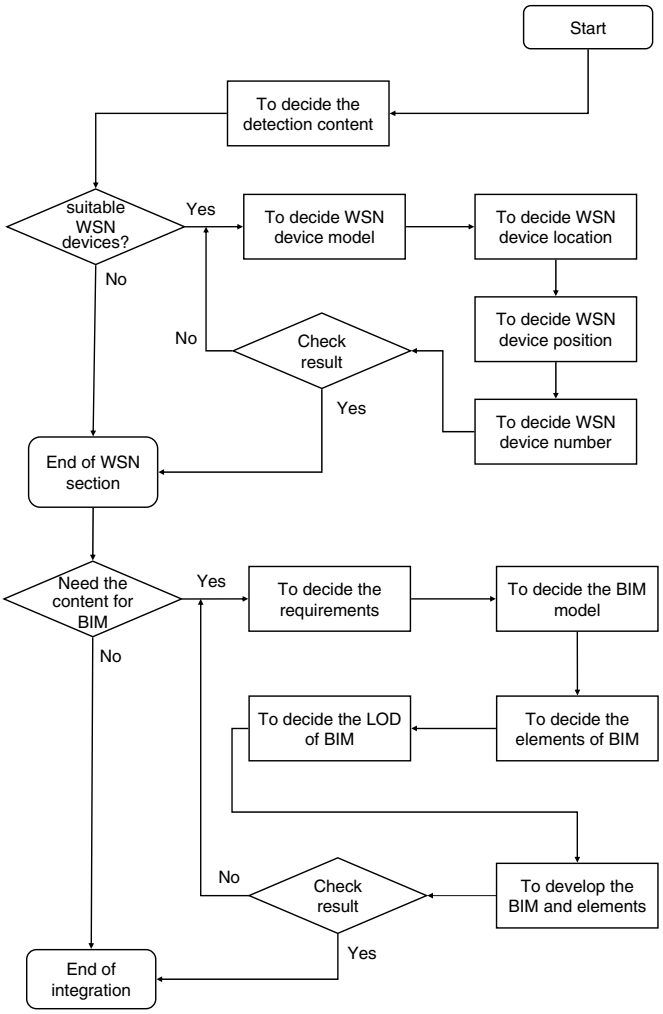


Fig. 2. Workflow of proposed approach for integration of WSN and BIM technologies.

BIM for parking garages of the smart cities. First, a WSN monitoring system is established for collecting the CO gas, temperature, and humidity status, with many sensor nodes placed in an underground garage. The data is then collected and the locational status is displayed on a BIM model, which actively shows the overall environmental and safety statuses on the DT with color-coding corresponding to the current conditions. With regard to safety concerns, WSN nodes with a smart control function are also designed which can automatically activate the ventilator and flashing alarm to remove hazards and give a safety warning when a high concentration of CO gas is detected. Fig. 3 shows the concept of the proposed system. Furthermore, a multilayer architecture is developed to clarify the overall designed framework, consisting of (1) a presentation layer, (2) an application layer, (3) a database layer, and (4) a sensor layer. The entities and information of different attributes are divided into different layers, with the relationships and data flow between various subsystems and entities identified. Fig. 4 shows the overall architecture and data flow relations between layers, and the descriptions are given in Table 1.

System Design and Development

Both subsystems, WSN and BIM, are proposed and comprised to establish the DT system in the study. The WSN system was constituted by many sensor nodes with various functions, with each node being an independent device responsible for individual duties such as environmental condition sensing, data communication, and hardware control. Every node is designed to have a unique function according to the required duty in the WSN system; the interaction of various functions enables the intelligent capability of the system. The other subsystem, BIM (DT), is a digital 3D model of the building basement composed of different components which display spatial information, material contents, and observed conditions.

When the system starts, the location-specific conditions of the garage, including gas concentrations, temperature, and humidity, will be monitored and transferred to the host by WSN node relays. Then the BIM components corresponding to the monitoring

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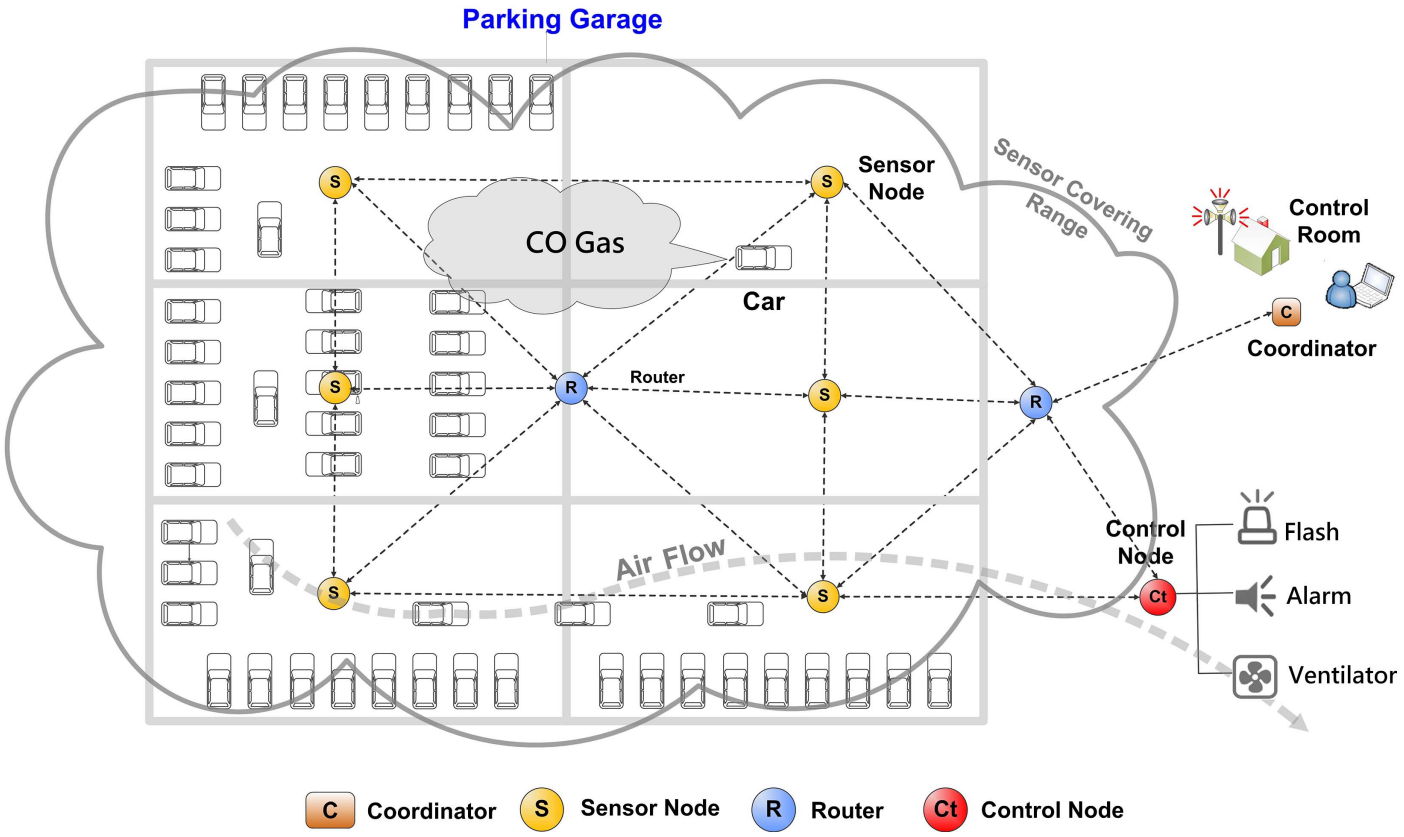


Fig. 3. Multilayer analysis and data flow process.

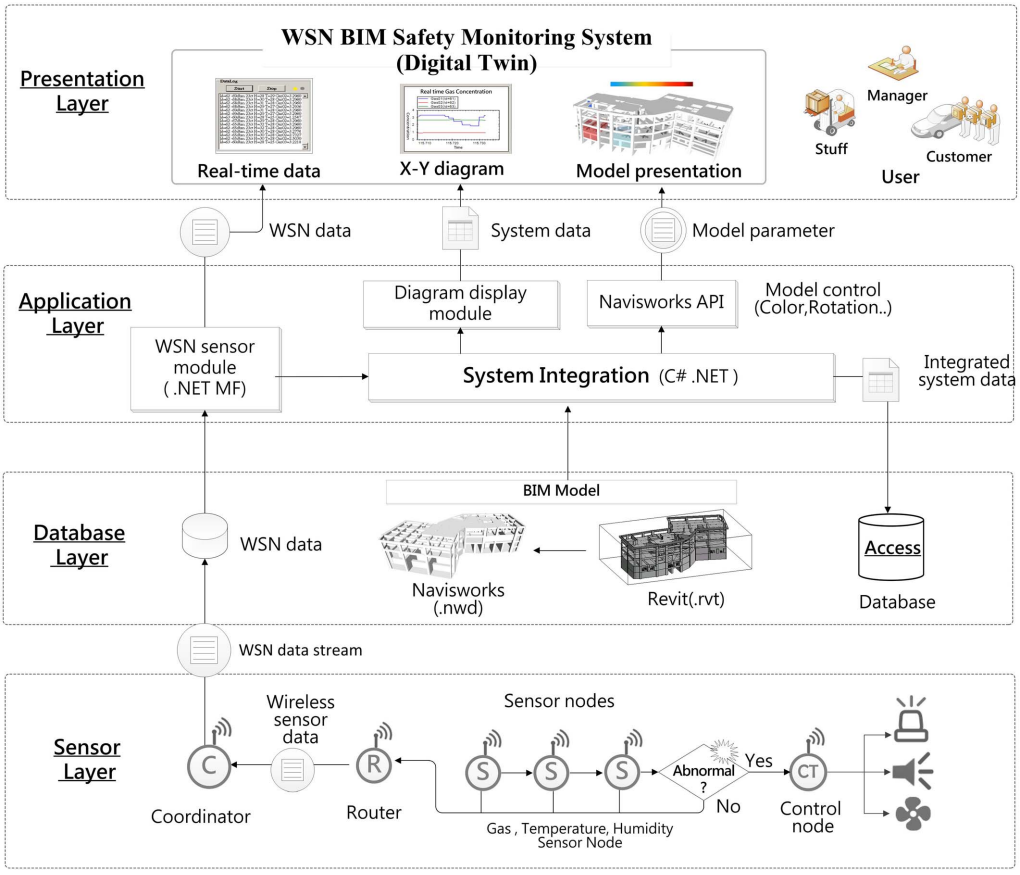


Fig. 4. Multilayer analysis and data flow process.

Table 1. Description and tasks of designed layers

Layer	Description	Task
1. Presentation layer	1. Present the collected information in real time	1. To show the real-time gas value and drawing curve on specific window
	2. A dynamic model (DT) for representing physic condition and management	2. To establish a BIM model within active color display capabilities
	3. User interface design for efficient management	3. To design an optimal user interface of the DT system for monitoring
2. Application layer	1. Design the WSN nodes and functions	1. To choose suitable software and methods to achieve the design functions
	2. Establish the BIM model and modification	2. To arrange the data flow between applications (WSN, BIM model, and system)
	3. Combination of WSN, model, and information for the DT system	
3. Database layer	1. Database for storage of sensor and system data	1. To establish the data storage of historical recording and processing
	2. Management database and historical information inquiry	2. To build a digital model corresponding with building, assets, and locations
	3. Manage the spatial and conditional information of DT	3. To choose the BIM components for DT condition representation
4. Sensor layer	1. WSN system plan for underground parking garage detection	1. To design the WSN nodes with various functions
	2. Development the sensor nodes, including: (a) sensor node (b) router (c) control node, and (d) coordinator	2. To construct the Zigbee-based WSN nodes and network 3. To communicate the gas and condition information in a garage environment 4. To develop the WSN smart functions for actively monitoring and exhausting CO gas and power saving
	3. Intelligent function design for maintaining environmental quality and safety	

Table 2. The tasks of developing phase and adopted applications

Task	Application/software	Description
1. Development of functional WSN nodes and network	• .NET microframework	1. Sensor node: comprises CO gas, temperature, and humidity components and Zigbee module and the corresponding capability in the detection and data communication 2. Coordinator: hosts the returned data from the sensor node and transfers to the computer 3. Control node: actively switch on safety devices when anomalies are detected 4. Router: relay WSN signal and extend transmission distance
2. Establishment of BIM model and modification	• Autodesk Revit • Autodesk Navisworks • Navisworks API	1. Establish original BIM model for DT by Revit (.rvt) 2. Transfer .rvt file to Navisworks (.nwd) file for monitoring and display 3. Control the component color change and model actions by Navisworks API
3. Processing of data	• C# (.NET Framework) • Access, Excel	1. Treat the collected data and deliver to Access and Excel applications for storage and processing 2. Show real-time X-Y curve diagrams of gas, temperature, and humidity conditions
4. Combination of BIM and WSN	• C# (.NET Framework) • Navisworks API	1. Manage the actions of BIM model (move, rotate, cut, zoom in/out, etc.) 2. Create BIM components for conditions integration and display (gas, temperature, and humidity) 3. Change the components color display and alert the emergent areas
5. Environmental and safety information management system	• C#(.NET Framework) • Access	1. Manage historical data storage 2. Check detail duration data of emergency 3. Manage the registration data of sensor nodes

locations will dynamically display the color and indicate the gas and environmental conditions according to the returned information. The overall conditions of the monitoring locations can be clearly displayed and mastered on the active DT model. The tasks of developing phase include (1) designing and developing the proposed WSN sensor nodes and network system, (2) developing the BIM model of the physical garage environment, (3) integrating the subsystems, (4) designing the database and data processing, and (5) implementing and testing a prototype system. Table 2 lists the task items, applications, and descriptions, and Fig. 5 demonstrates the relationship of integrating subsystems and applications.

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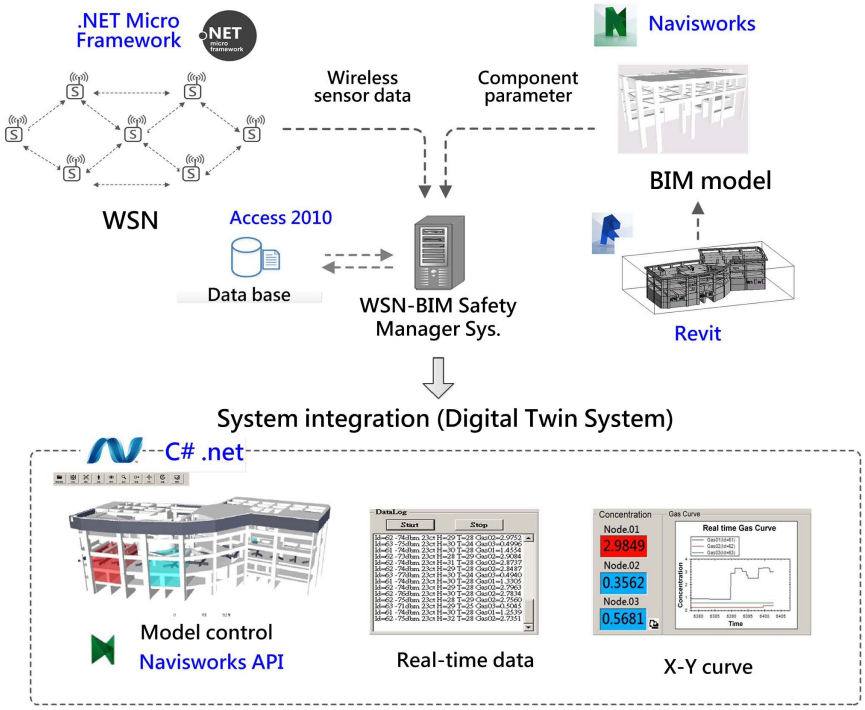


Fig. 5. Adopted applications and integrating relationship.

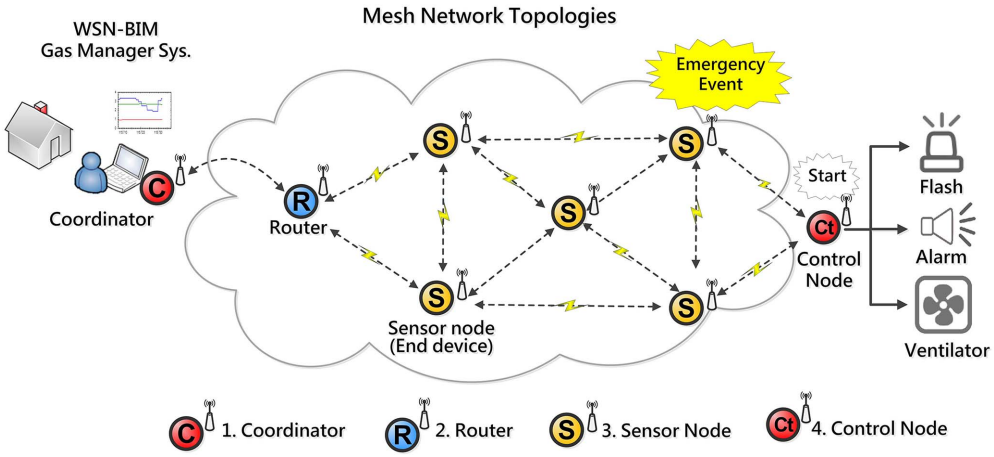


Fig. 6. Proposed mesh topology of network system.

WSN Topologies and Nodes Development

Topology Design

The WSN provides the DT system with information for updating the status in real time. The WSN is configured by a host and sensor nodes. In general, for a typical WSN, topologies can be classified as *star*, *tree*, and *mesh* (Farahani 2018). In the star topology, each node is connected with the coordinator (host) individually and the data is transferred directly to the host with no node relay, therefore resulting in short and restrictive communication distances. The tree topology is comprised of several star networks in a clustered arrangement, but the communication may be disconnected if the cluster-head node has failed. In a mesh topology network, every node cooperates with each other; the communication can be extended to a longer distance by ad hoc node relays of nodes. In addition, the mesh topology can recover the network automatically by self-connecting capabilities if any internal node has failed. In this study, mesh topology was designed as the backbone of network, as it can provide a longer communication distance and data transfer stability (Fig. 6).

Developing Functional WSN Nodes

The hardware framework of the sensor node was constructed by the Microsoft .NET Gadgeteer embedded system and the functions were designed by the .NET Micro Framework of Microsoft (.NET MF). The WSN system is comprised of different functional nodes. In this study, four types of WSN nodes were developed, including the *coordinator*, *sensors node*, *router*, and *control node*. The coordinator is connected with a computer and acts as a host, which is in charge of launching the network and receives the data sent

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Table 3. Developed WSN nodes

Type	Function
1. Coordinator	1. To launch a WSN network and coordinate 2. To act as the host and collect data sent by all sensor nodes
2. Router	1. To transfer and relay the signal from parental WSN node 2. To expand communication range and signal coverage
3. Sensor node	1. To detect the gas, temperature, and humidity conditions of the garage 2. To judge the urgent situation and inform the control node to switch on safety devices 3. To increase the data logging frequency for recording the detail data of emergency duration
4. Control node	1. To switch on/off the safety devices (flashing alarm and ventilator) for automatically removing hazards and risk reduction

by all sensor nodes. The router, without equipping any sensor, only transfers the data to the next node and extends the transmission distance. The sensor node is connected with CO-gas-, temperature-, and humidity-sensing components and is responsible for monitoring and transmitting the measured data back to the host. Additionally, the control node is equipped with an electric relay that enables the node to switch connected equipment on/off for special purposes. In this study, intelligent WSN functions were designed so that when any of the nodes in the network measured the urgent status, the control node will immediately switch on both the ventilator (to exhaust hazardous gas) and the flashing alarm (to alert others of danger). The developed WSN node types are described in Table 3.

Composition of the Sensor Nodes

The sensor node is usually the most numerous and important device in monitoring. To enable the node to sense gas and environmental conditions and communicate the information, the following sensor components and modules were designed (Fig. 7):

1. CO gas sensor: The MQ-7 CO gas-sensing component (Hanwei Eletronics 2019) is equipped to detect CO gas and transfer a proportional direct current output.
2. Temperature- and humidity-detecting module: The DHT-11(Mouser Electronics 2018) module was adopted to detect the temperature and humidity conditions.
3. Wireless module: The Zigbee CC2530 wireless module (Texas Instruments 2019) was selected to construct the WSN backbone, and the mesh topology was developed.
4. Main chip board: The Microsoft .NET Gadgeteer board with an ARM Cortex-M4 processor provides the hardware platform for the embedded system.
5. Power supply module: The power supply comprises 18,650 lithium ion batteries that provide the operating power source for the nodes.

The node mainboard comprises the components by cable lines, and during operation the environmental condition data is measured and then transferred by the Zigbee module. The data recording frequency of the nodes was programmed to routinely operate at a low frequency (0.2 Hz) for power saving; when the emergent situation is detected, the frequency will increase to 3 Hz automatically, as the information is important for future investigations. Meanwhile, the sensor node will judge the abnormal condition smartly and inform the control node to switch on safety devices, which reduces the risk posed by the accumulating CO gas and enhances human safety. Fig. 8 presents the prototype of the sensor node, and Fig. 9 demonstrates the programming flowchart. All of the developed WSN nodes are presented in Fig. 10.

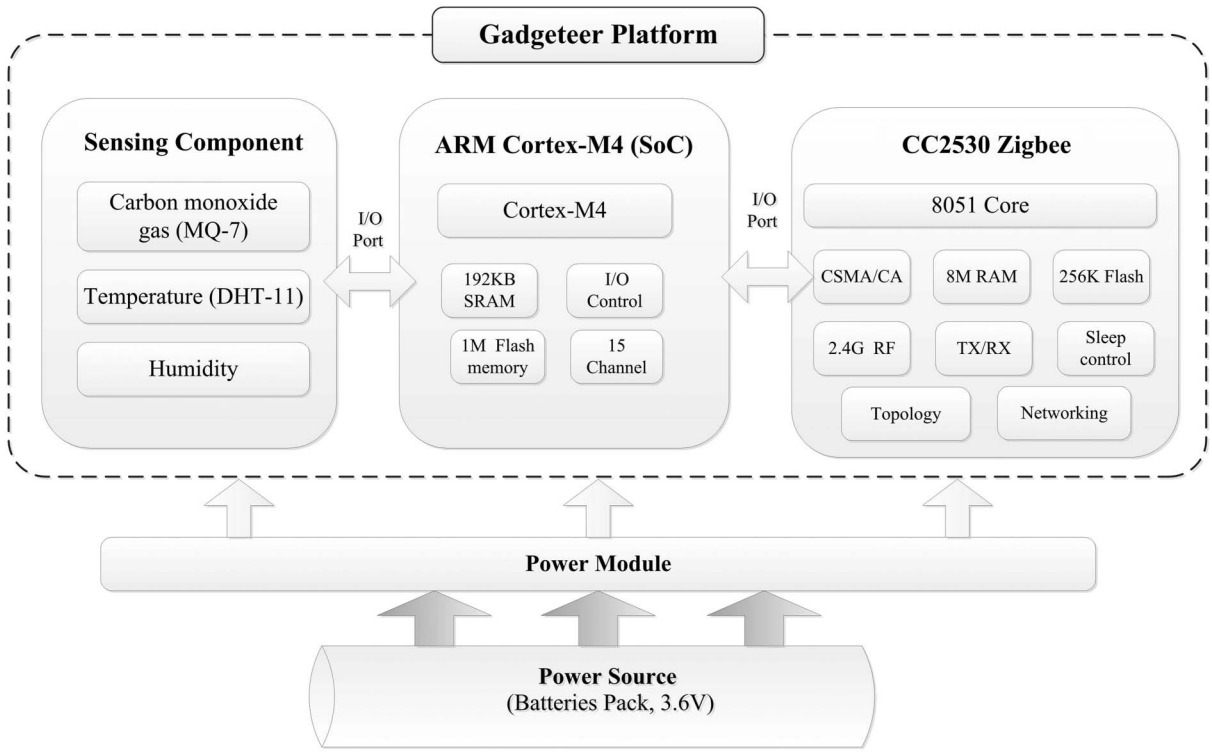


Fig. 7. Designed composition of wireless sensor node.

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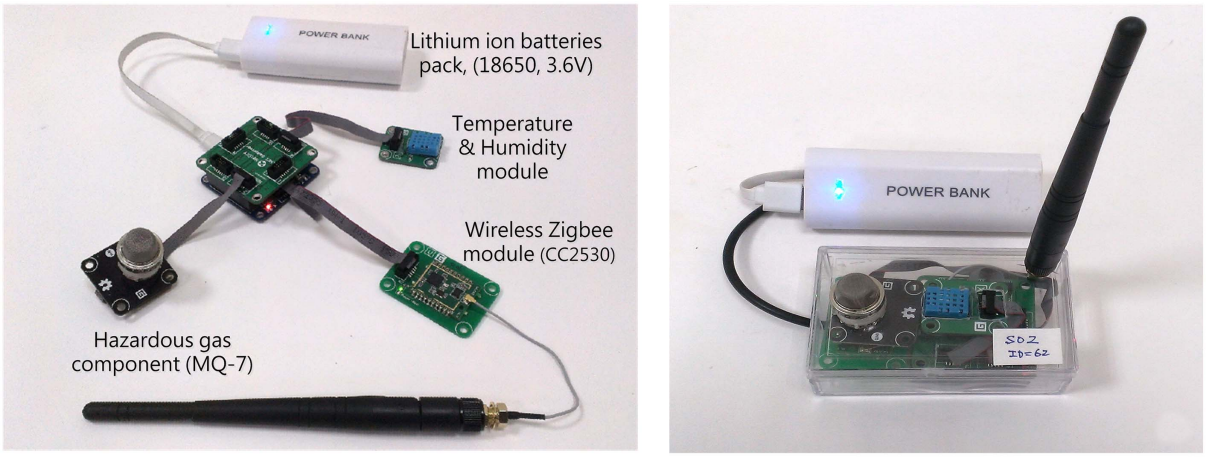


Fig. 8. Prototype of the sensor node: (a) dispread; and (b) integrated.

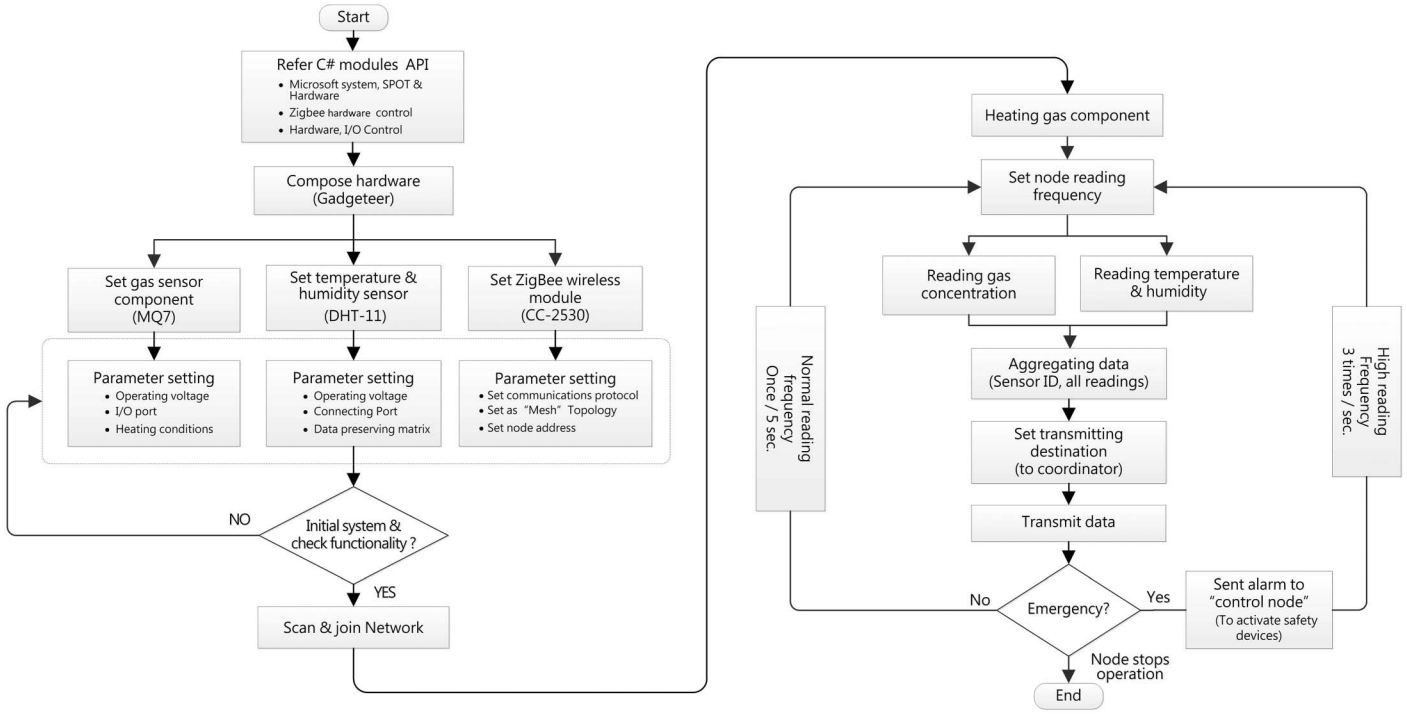


Fig. 9. Design flow of sensor node function.

Build and Command BIM Model

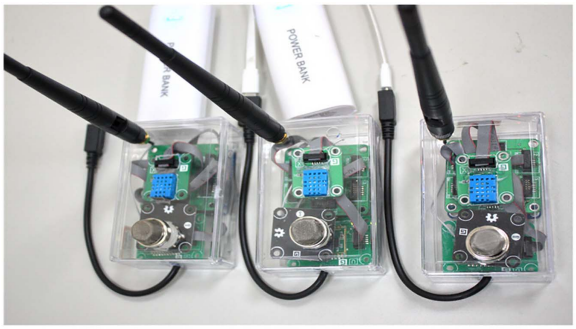
For the purpose of the DT development, the BIM model of the garage site must first be established. The model provides a digital and visual representation for the application of the DT to monitoring real-time location-specific conditions. The CO gas concentration, temperature, and humidity conditions collected by the sensor nodes are indicated by dynamic color-coded model components. The risk status of the monitoring location, varying from high to low, is indicated by a red-to-blue color gradient according to the returned CO gas concentration. The comfort level in terms of temperature and humidity is also indicated by the green-to-red color gradient. Hence, both the safety and environmental information at various monitoring locations can be displayed simultaneously. The real-time active BIM model provides a virtual DT function to represent the actual safety and comfort level in the monitored garage, which also greatly enhances the efficiency of management of the quality of the environment.

Build the BIM Model

The DT model of the garage was built by Autodesk Revit software (.rvt file) and transferred to a Navisworks (.nwd) file, which retains the original assets, attribution, and geometrical information and can be controlled by an application programming interface (API) program. In practice, the outside shield of the BIM model structure should be removed or modified, so that when operating the model, the internal assets and conditions can be displayed. Moreover, for the purpose of displaying various conditions, room components

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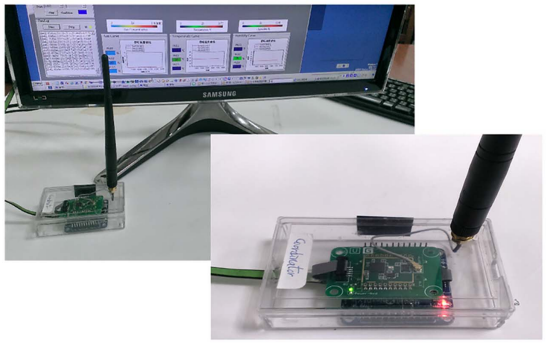
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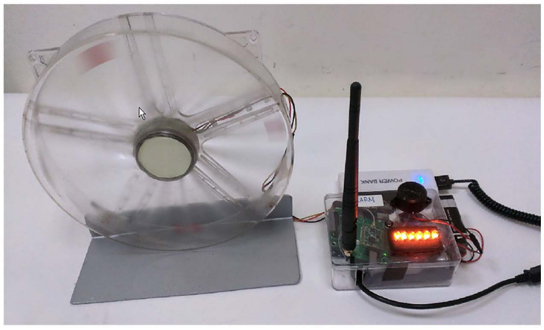
Sensor Node



Router



Coordinator
(connect with PC)



Control Node
(connect with flash alarm & ventilator)

Fig. 10. Four prototypes of the developed WSN nodes.

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corresponding to the monitoring locations were created. For example, the added component for gas concentration was modified to display the real-time CO gas concentration by color-coded indication. The same method is also proposed to indicate the temperature and humidity statuses (Fig. 11).

Command BIM Model and Condition Display

APIs are functional interface programs provided by application supplier which enable communication between different applications, and enable the developer to design a customized system. The Navisworks API was applied to control the BIM component's external attribution (e.g., color or transparency) and enable the model to be controlled (e.g., rotation, move/pan, and zoom up/down function). The API was used to control and change the model color for indication and to develop an efficient interface for operation by the user.

When operating the system, the data stream feedback from the WSN nodes is collected through the coordinator, with every data set containing both the spatial information and environmental conditions. The former is used to determine where the data originates and thus select the corresponding component for display. The latter consist of the gas concentration, temperature, and humidity values, which will be processed to indicate the relative status via the color. Each component will dynamically change color and indicate the locational status according to the incoming data. The integration mechanism between the streaming data and the model component is analyzed in Fig. 12.

System Integration

The developed DT system comprises three categories of information, including the BIM model, WSN, and database, as well as integrated and C# applications. The system also has an optimized user interface: in addition to an active BIM model for the integration of location-specific conditions, the measured gas concentration, temperature, and humidity values and relative X-Y curve diagram are also displayed on a specific window in real time. When the system is activated, the coordinator launches the network first. The other nodes later join the network in sequence and begin to automatically return data. The data is sent from sensor nodes in various measuring locations in the garage, communicated through the node relay. Each data set is then checked to confirm which location the data is from, and then the relative component color will be displayed corresponding to the location and measured condition. Moreover, the control commands of BIM enable the model to be moved, rotated, cut, zoomed in/out, or adjusted for the best view sight/angle for monitoring by the user (Fig. 13). The measured data, including the transferring time, node address, signal strength, gas concentration, temperature, and humidity, is then stored on a Microsoft Access database for long-term observation and tracking. A screenshot of the integrated DT system is shown in Fig. 14.

Case Study

Description of the Case Study

An underground parking garage of a depot located in northern Taiwan was selected as a case study. The garage has an area of about 50,000 square meters with two underground floors. The air is heavily polluted by cars and motorcycles during the rush hour, and thus the drivers and passengers and working staff may be exposed to hazardous gases and possibly become injured.

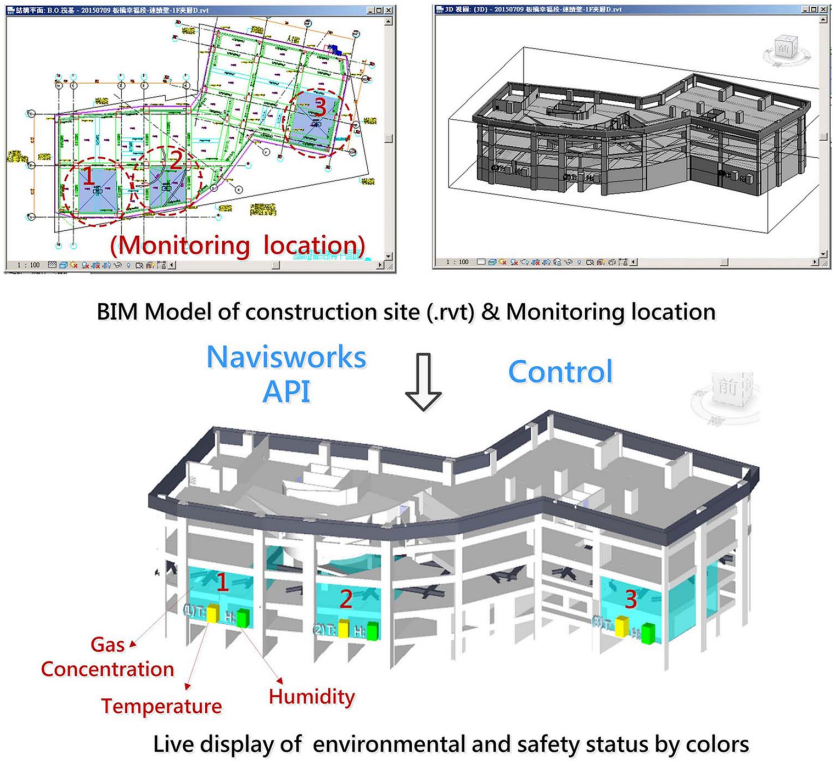


Fig. 11. Establishment of the BIM model and indicating components.

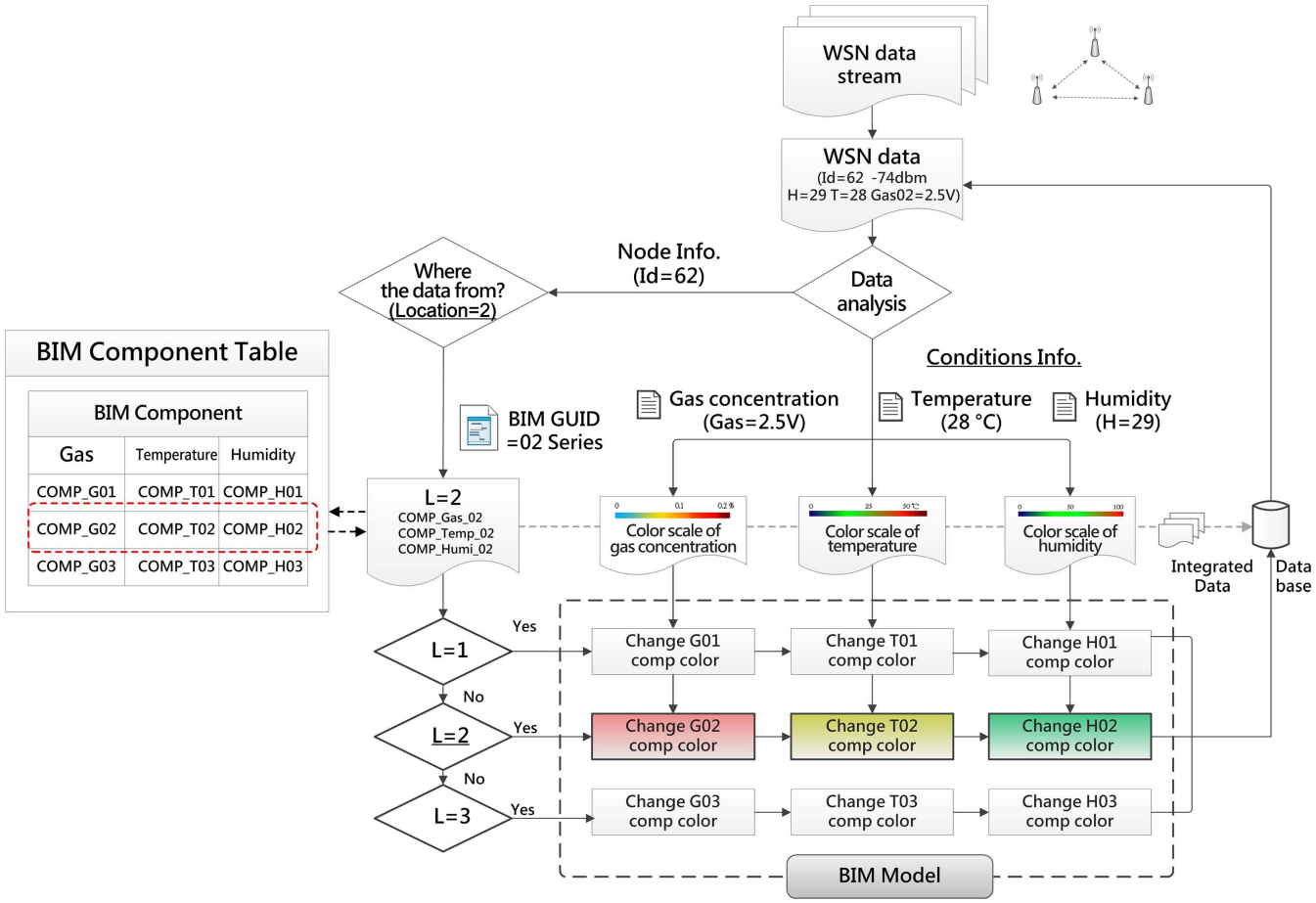


Fig. 12. Integration mechanism of streaming data and model component.

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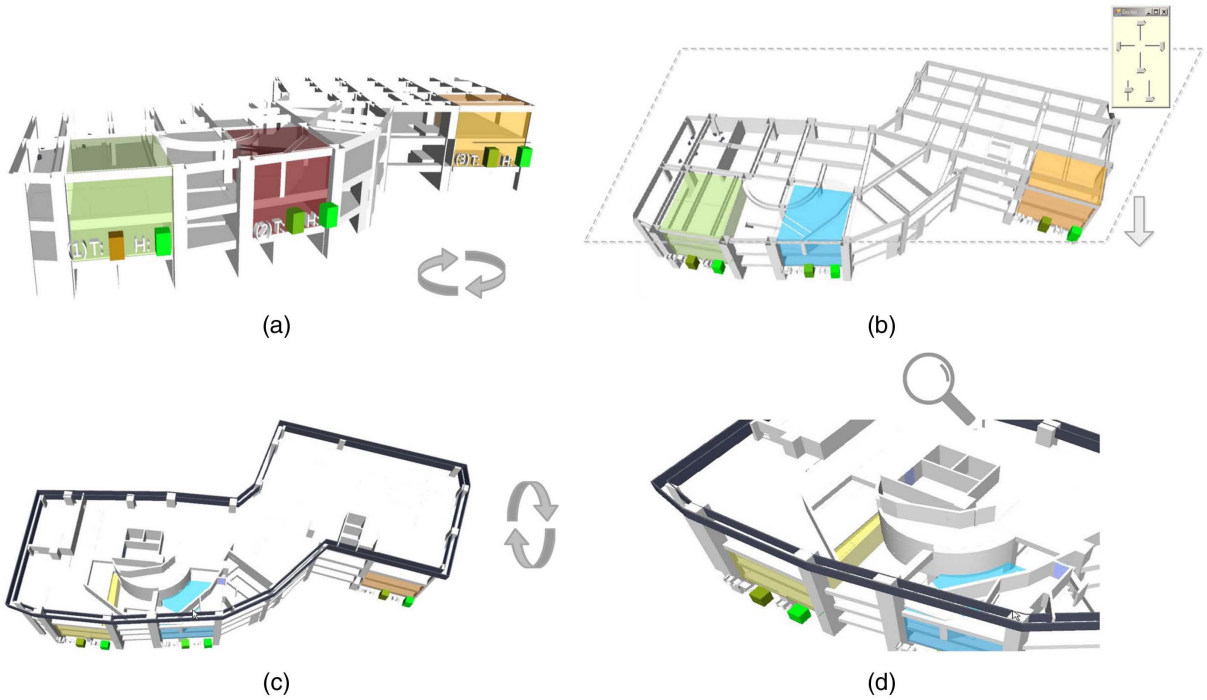


Fig. 13. Multiview control of the live conditions on BIM model: (a) overall view and rotate; (b) cut model (from top); (c) flip model; and (d) zoom in/out.

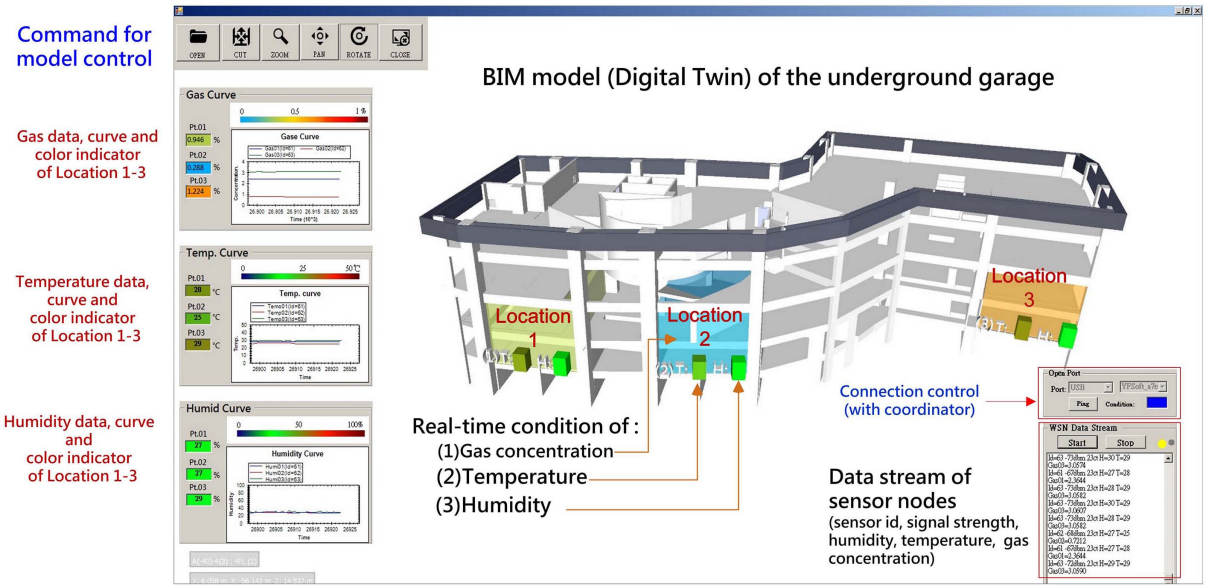


Fig. 14. Overview of the developing DT system.

In the test, three locations in the garage were chosen to be monitored, with three sensor nodes set up to measure the CO gas concentration, temperature, and humidity conditions. The coordinator (host) was connected to a computer and placed near the garage entrance on the ground. Two routers were set up between the sensor nodes and coordinator for linking the data communication so that the sensor data from the different garage locations could be collected and transferred to the host. Additionally, a control node was set up near the air conditioning vent, which was connected with safety devices (ventilator and flashing alarm) so that when any abnormal CO gas conditions were detected, the safety devices would activate immediately to remove hazards and warn persons to evacuate.

A BIM model of the building and garage was established in advance and set as the DT for the system monitoring. Three locations were assigned as monitoring positions, and three components were established to indicate the conditions (gas, temperature, and humidity) for each position.

During the experimental process, the nodes of the WSN created a network automatically and operated via an independent power source. The communication route of the sensor nodes was planned

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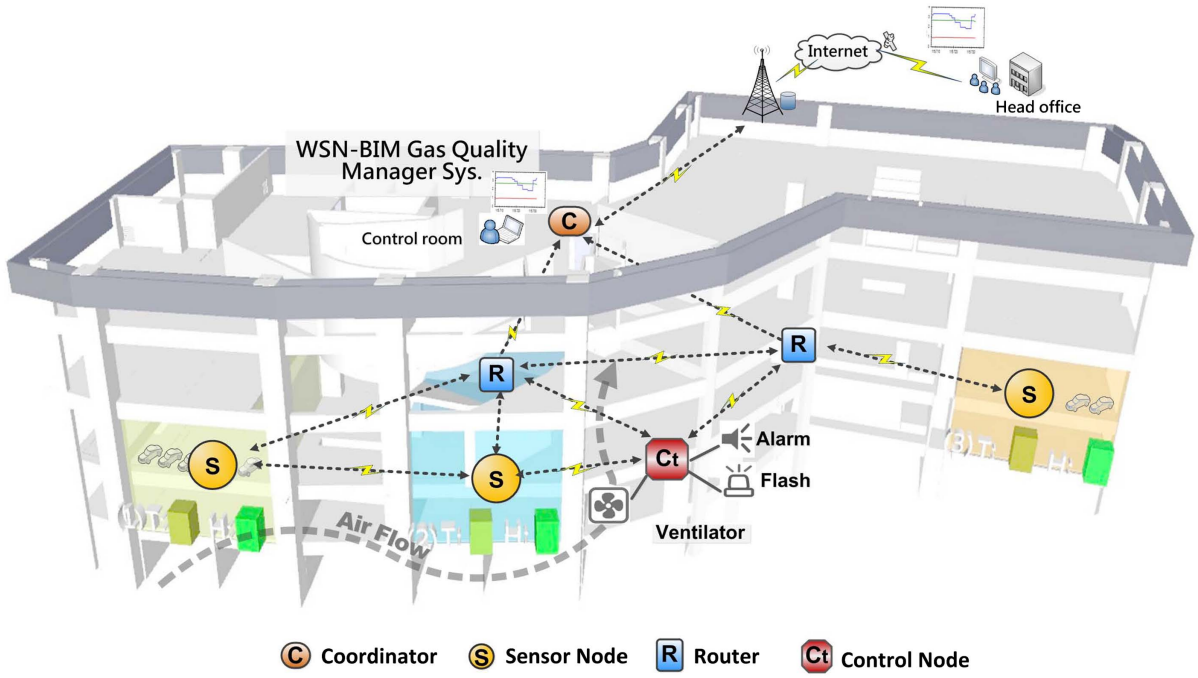


Fig. 15. Layout of the WSN experiment in a parking garage.

Table 4. Test items of the case study

Items	Description
1. Capabilities of hazardous gas detection	1. To set up the WSN network by three sensor nodes and a control node connecting with safety devices 2. To simulate the emergent situation of high concentrations of CO gas occurring and to check the system reaction 3. To test the designed intelligent function (to increase recording frequency and to actuate the ventilator and flashing alarm)
2. Communication distance and router extension test	1. To test the longest communication distance and signal attenuation between sensor nodes 2. The distance extension by router installation
3. System interference	1. To test the signal interference by placing near an operating machine (such as a high-speed ventilator)

to be a conservatively estimated distance of about 200 m. If the signal could not be transmitted, a router was added on the route between two nodes to extend the signal. The layout of the test location is shown in Fig. 15.

Three operational elements were tested in the case study: (1) the system capabilities in hazardous gas detection, (2) communication distance and a router extension test, and (3) system interference. The elements are described in Table 4.

Performance Test of Gas Detection

This test checks the system performance by simulating the occurrence of a hazardous gas. First, a notebook installed the DT system of the parking garage model and was linked to a coordinator for receiving the detection data from the sensor nodes. Then, the WSN network was established by the coordinator and the joined WSN nodes. Staff moved the sensor nodes to different monitoring locations in the garage and provided feedback by handheld radio when the sensor nodes were set up. In the next step, the combustible gas was emitted by a lighter held close to the sensor node, which simulated the emission of hazardous gases so as to check whether the system could immediately react to the emergency situation (Fig. 16).

The experimental results displayed that the proposed system reacted to the hazardous gas in approximately 1–2 s and a red-color warning immediately indicated the abnormal areas. Meanwhile, the control node trigged the ventilator and flashing alarm automatically, which eliminated the hazardous gas accumulation and prevented nearby persons from inhalation injuries. The results demonstrated that the system could execute the monitoring and removal tasks for hazardous gases, significantly reducing the risk in parking garage management (Fig. 17).

Signal Transmission Test

A limitation in communication was introduced to test the longest transfer distance of the WSN signal. The sensor node was carried by staff and moved directly in a straight line from the coordinator. The staff moved and reported the signal strength every 20 m until the signal became weak and disappeared. The test results demonstrate that the maximum signal communication distance in a straight line was approximately 250 m between two nodes, with the signal strength decreasing approximately 0.3 dBm per meter. However, if a router node was installed between the sensor nodes, the signal would then be relayed, resulting in the transmission distance increasing to more than 450 m (Fig. 18).

Interference Test

This experiment was designed to check the WSN signal interference from operating equipment, (e.g., the high-speed running ventilator or motor). A running sensor node with a communicating signal was set on top of a high-speed operating air conditioner. The signal stability and strength were observed and compared with the

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Fig. 16. WSN experiment in an underground parking garage.

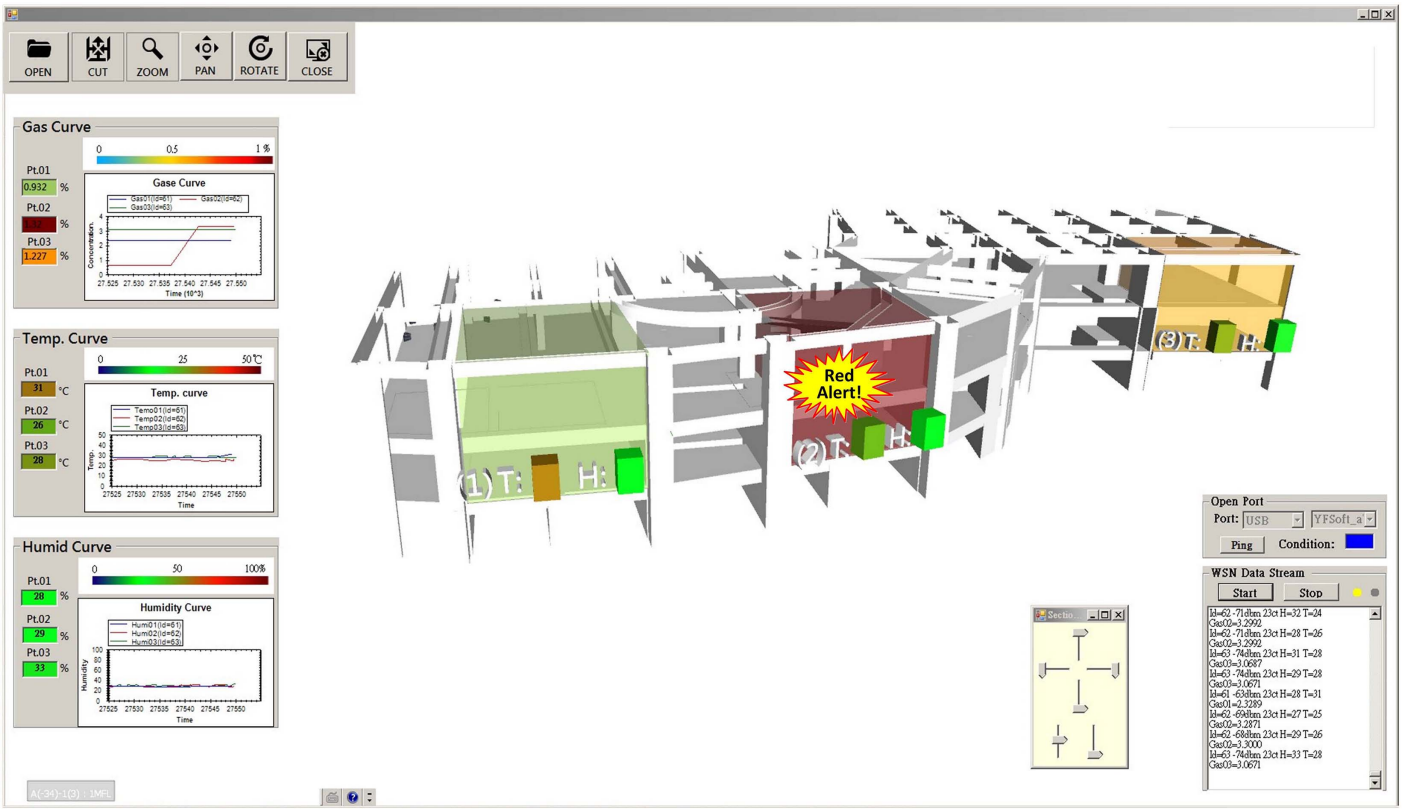


Fig. 17. Screenshot of the operating DT system in the experiment. (The model has been cut to the monitoring layer for best view; emergent gas status was detected at Location 2 and warned by red.)

baseline case. The results indicate that only minor interference in the stability and signal strength of the WSN resulted from the high-speed running equipment (Fig. 19).

Evaluation and Discussion

This paper proposed a DT application for garage safety and environmental quality management via integration of WSN and BIM technologies. The proposed system was tested and demonstrates the applicability in underground parking garages. The function and performance were tested to determine the capabilities of the system. The proposed system provides an advanced method to enhance the environmental quality and safety as well as decrease the power consumption and operating costs. The proposed method can be applied to other infrastructure management applications.

In the case study, the WSN communication distance was 250 m. However, the signal became attenuated when the communication route was nonlinear. The test results displayed an approximate 15%–20% loss of signal strength after a 90° turn, with similar results for both horizontal and vertical directions. Additionally, the WSN signal can pass through a concrete wall, although the signal strength will decrease depending on the material. In the test case,

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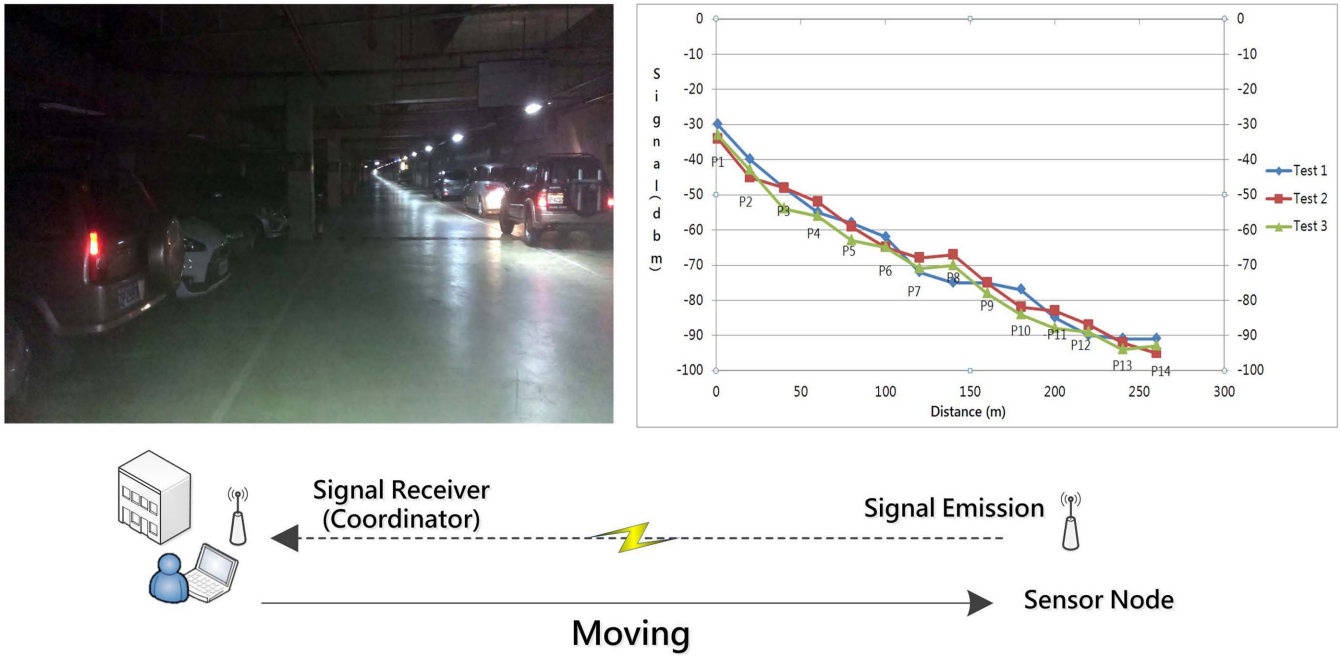


Fig. 18. Performance curve of WSN in lane communication test.



Test item	Original (dBm)	Test Result (dBm)	Estimation
(1) Interference by high speed air cooling fan	-55~-56	-54~-57	unapparent effect
(2) Interference by a car engine	-56~-58	-55~-58	unapparent effect

Fig. 19. Interference experiment of sensor node.

a 30% ~ 40% loss in signal strength was observed after passing through a 25-cm-thick hollow brick wall. These results may be used when planning the layout of the sensor node. If the achievable communication distance is less than the distance to the next node, installation of a router is essential for extending the data communication distance.

The installation of a WSN is easy and convenient. Just switch on the power of the node and place it in the desired measuring location. The node will link with the WSN network and begin to return data automatically, demonstrating flexibility and applicability.

In the duration assessment, the WSN power source was either batteries packs or an external power supply. If the node is installed on a moving object or creature or the external power is not available, then individual batteries packs are required. In this study, the power supplied by six 18,650 lithium batteries packs enabled the WSN node to operate for approximately 1 week. An extra power source is needed for longer operation, and thus the power supply and battery capacity are important concerns in WSN operation.

In view of the cost assessment, the cost for each of the proposed nodes is about \$80–\$100, and a basic measurement (single point detection) requires a sensor node and the coordinator, which costs nearly \$200 (not including the price of a computer). If the detection point number is increased, the additional cost is \$100 for each node. When compared with the most commonly used detector (the handheld detector, which has a price ranging from \$200 to \$400 each), if four locations are desired to be simultaneously monitored, the total cost of the handheld detectors comes to \$800 (\$200 each), with additional cost for human operation. If the proposed WSN detection system is used instead, the cost is reduced to approximately \$500 (a coordinator with four sensor nodes), resulting in nearly 40% cost savings as compared to the handheld detectors.

In view of expandability, the proposed WSN system involves incorporation of individual sensing components and a wireless communication module. The design enables the node to be equipped with different types of sensing components as required and yet still operate within the original wireless backbone. The detection function can be easily modified or expanded for application to different task requirements.

The following suggestions are summarized based on the study results:

1. It is very important to identify the requirements and develop the initial plan according to the requirements. When the final as-built BIM model is developed at the end of construction phase, it is necessary to understand and decide which elements of the model will be decreased for environmental monitoring

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management. All elements of the BIM model should be removed or decreased if they are not used or applied during the monitoring processes. The initial consideration will be helpful in enhancing the operation performance of the proposed DT monitoring system.

2. Given that the gas sensor output reading is a serial voltage change corresponding to the gas concentration, for quantification of the actual gas concentration, the calibration of the sensor by standard gases is recommended, as the actual concentration is meaningful in hazardous gas control and safety management.
3. Battery capacity and operation duration were the major limitations of the WSN operation. In particular, the gas sensor has to work in higher temperature environments for gas detection, which will require more electricity. It is recommended to adopt high-capacity batteries or a fixed-type power source for long-term operation.
4. Considering the requirements of parking garage environment management, the mesh network topology which provides extended communication distance and network reconnecting abilities is advantageous for the monitoring task; star and tree topologies may disconnect and lose data in the case of far-distance communication.
5. The system provides various information, including hazardous gas concentration and environmental conditions, and their relative locations simultaneously, so the risk priority of the locations can be estimated for reference and further calculated as a localization algorithm to support the evacuation plan in a fire emergency.
6. In consideration of expandability and increasing the system value, the functions of the sensor nodes can be increased to include free parking space detection by the addition of an infrared sensor, which could support the parking space guide and the parking management system.

Conclusions

Currently, there are many related technologies suitable for application of DT for smart cities. Specially, the assistance of BIM technology can provide visual 3D communication for DT. Furthermore, it is easy to obtain the required information from the devices located in the selected areas using WSN technologies for IoT application in smart cities. Although there are some previous research or practice case studies using BIM and WSN technologies, few studies focus on the approach for the integration of WSN and BIM technologies for environmental monitoring management in parking garages of smart cities. The proposed approach in the study has major differences and contributions compared with previous researches.

This study proposed an advanced monitoring and control system for underground garage environment management adopting WSN and BIM technologies. The WSN acts as a nervous system for hazards detection, working to maintain the safety of the garage environment. The collected WSN data enables the BIM model to actively reflect the current state in the garage and provide a visually dynamic interface for management. The integrated system provides an efficient platform in garage environment monitoring while maintaining safety. This model can also be easily transplanted to other monitoring systems for further application.

In analysis of the interoperability of the integrating system of WSN and BIM, the WSN provides the active information measured from the real world and enables the BIM model to be a real-time DT system and show the real status for management; on the other hand, the BIM provides a digital structure for the WSN data to be correctly embedded and shows the dimensional relation of the monitoring position for observation and judgement. Moreover, the color gradient displays the risk degree of multiple locations, so that the risk priority of the monitoring locations can be visually displayed by the model to provide an awareness of risk distribution of the whole space, which enhances the safety management, emergency response, and decision-making.

The DT may contain many assets of different sizes which are closely distributed on the model and may obstruct each other. Additionally, many conditions and statuses have to display on the model area simultaneously, increasing observation obstruction. How to properly arrange the layout will be an important concern in system design. Furthermore, the proposed system achieves remote, real-time, and multipoint conditions monitoring. Meanwhile, the spatial relation and environmental condition information is provided simultaneously, and the observation and analysis of gas concentration and flow status can provide the information needed to determine the priority or the key position of garage air pollution alleviation mechanisms. The designed smart functions enable the system to start the ventilation and safety devices automatically when high concentrations of CO gases are detected, which can significantly enhance the safety of the garage environment and eliminate human injuries. Additionally, the operating and maintenance costs of the garage are also reduced. All the conditions are dynamically displayed by corresponding spatial components and indicate the location risk status by changing colors. The real-time information can be used for regular management and emergency response. Finally, the intelligent function of the WSN system was designed to remove hazards automatically when detecting abnormal situations. In the future, more related IoT technologies for underground garage environment management can be adopted and improved for advanced control and management. Furthermore, the applications of BIM can be integrated with artificial intelligence (AI) technologies for visually effective smart environment management in the public area of smart cities.

Data Availability Statement

All data, models, or code generated or used during the study are proprietary or confidential in nature and may only be provided with restrictions.

Acknowledgments

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